HEXLEY CHEMICALS

Permit Standard: Hot Work

Document Standard 01 | Issue 7 | Owner: Process Safety | Review cycle: every twelve months

1. Purpose and scope

This standard sets out what a request for its permit type must show before an Area Authority may issue the permit, and what follows the decision.

It is used by the Permit Coordinator or Area Authority who opens a review, by the Area Authority who decides it, and by the performing authority who requests the permit, as the Control of Work Policy describes.

Each permit type has its own permit standard. The five permit standards share one layout and many of the same sentences; they differ in what the permit type covers, in the requirements of section 4 and how section 5 reads them, and in the documentation, the writing that follows a decision and the reasons that sections 6 to 9 give. A request is always reviewed under the standard of its own permit type, and under the standard of its linked permit type where it has one; a requirement in one standard never applies to a request of another permit type unless the request is linked to it.

The standard applies to every request for its permit type, whichever channel it comes through.

2. What this permit covers

This standard covers hot work anywhere at Thornmere Works: welding, grinding, flame cutting, brazing and hot tapping, and any other work that makes a flame, sparks or hot metal.

Hot work inside a confined space needs a confined space entry permit too, linked to this one, and hot work done at height needs a work at height permit. The hot work and confined space entry standards are written from one template and read almost alike; their requirements are not the same.

The permit type is Hot work.

3. Who requests

A request is made by the performing authority - the contractor's supervisor who will lead the job - through the channel the contractor works in: the site portal, the contractor app or scanned paper. The performing authority answers for the request and the documents attached to it.

The request gives the plant area, its hazardous area classification, whether the process unit there is operating or shut down, whether the work is indoors or outdoors, the permit type and any linked permit type, the work activity, the equipment tag, the size of the work party, the planned work hours and the answers its requirements read. The documents attached to it - a method statement, a task risk assessment, a gas test record or a rescue plan - describe the work.

A hot work request gives the gas readings at the work point, the clearance of combustible materials and the fire watch.

4. Requirements

Every requirement below must be Met for an Area Authority to issue the permit. Each requirement is stated once, in one sentence, and a request is screened against all of them, in this order:

1 **Risk assessment.** A task risk assessment for the job is attached to the request.

2 **Flammable gas.** The flammable gas reading at the work point is at most 5 percent of the lower explosive limit.

3 **Oxygen.** The oxygen reading is at least 19.5 percent and at most 23.5 percent.

4 **Combustibles cleared.** Combustible materials are cleared to at least 10 metres around the work.

5 **Fire watch.** A fire watch stays for at least 60 minutes after the work ends.

A requirement is never met by a sentence in a statement, and is never waived.

5. Reading the requirements

Each requirement is Met, Not met or Not documented on the request as the Permit Rules read it. The latest revision of a request is the request; an earlier revision is never screened.

- A task risk assessment is the contractor's own assessment of the job's hazards and the controls against them, attached to the request as one of its documents. The requirement is met when the request records it attached, not met when it records that it is not, and not documented when the answer is blank.
- The flammable gas reading is the gas test at the work point, in whole percent of the lower explosive limit, as the Permit Rules read it. A reading of exactly the figure meets the requirement, one above it fails it, and a blank reading is not documented.
- The oxygen reading is the gas test's oxygen in percent, to one decimal. A reading of exactly either figure meets the requirement, one below the lower figure or above the higher fails it, and a blank reading is not documented.
- The clearance is how far combustible materials are cleared around the work, in metres to one decimal, as the Permit Rules convert it from feet. A clearance of exactly the figure meets the requirement, a shorter one fails it, and a blank clearance is not documented.
- The fire watch is how long a person stays at the work area to watch for fire after the work ends, in whole minutes, as the Permit Rules convert it from hours. A fire watch of exactly the figure meets the requirement, a shorter one fails it, and a blank fire watch is not documented.

A requirement that is not documented names the answers it needs, so that the performing authority can give them.

A request with a linked permit type is screened under this standard and under the linked type's, and must meet both. The documents never decide a requirement. A gas testing sentence whose reading differs from the request's is flagged for the Area Authority, and the request's answer still decides.

6. Documentation to provide

When a request is returned to its requester, the performing authority must provide this documentation and send the request again with it: a task risk assessment for the job, the gas test record from the work point with its flammable gas and oxygen readings, the clearance of combustible materials measured around the work, and the fire watch arrangements with how long the fire watch stays after the work ends.

7. The permit document

The permit document goes to the performing authority when an Area Authority issues a permit under this standard. It names the request by its permit reference and permit type, says the permit is issued, and gives the plant area, the equipment tag, the work activity, the work party's size and the planned hours, with the flammable gas and oxygen readings at the work point, the clearance and the fire watch. A linked permit type is named with its own controls. It asks the performing authority to keep the gas monitor running at the work point and the fire watch in place for its full time. The permit document carries no figure from a model.

8. The work party briefing

The work party briefing is written after the permit document, for the people who will do the work. It tells them in plain words what the job is, where it is, its hazards and the controls they must keep. For hot work it says that the gas monitor stays on at the work point, that combustible materials stay cleared, and that the fire watch stays after the work ends. Where the permit depends on a linked permit, the briefing says so. The briefing never gives the suspension likelihood, the expected work hours or any other figure from a model, and never names any member of Hexley's staff.

9. Why these requirements

Hot work makes sparks, flames or hot metal that can ignite a flammable atmosphere or a combustible material. The flammable gas reading at the work point is therefore held to the figure that section 4 gives, far below the lower explosive limit, and the oxygen reading must lie in the band of normal air, because air enriched with oxygen makes everything burn faster.

Hexley holds hot work to a lower flammable gas reading than entry into a confined space, because hot work makes its own flame.

Sparks travel, so combustible materials are cleared to the distance that section 4 gives, and a fire watch stays after the work ends, because smouldering material can burst into flame long after the last weld.

Process Safety owns this standard, reviews it every twelve months and issues any amendment as a new issue of this document, replacing every earlier issue.